

Article

A Practical Image Augmentation Method for Construction Safety Using Object Range Expansion Synthesis

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Abstract: This study aims to propose a practical and realistic synthetic data generation method for object recognition in hazardous and data-scarce environments, such as construction sites. Artificial intelligence (AI) applications in such dynamic domains require domain-specific datasets, yet collecting real-world data can be challenging due to safety concerns, logistical constraints, and high labor costs. To address these limitations, we introduce object range expansion synthesis (ORES), a lightweight and non-generative method for generating synthetic image data by inserting real object masks into varied background scenes using open datasets. ORES synthesizes new scenes, while preserving scale and ground alignment, enabling controllable and realistic data augmentation. A dataset of 30,000 synthetic images was created using the proposed method and used to train an object recognition model. When tested on real-world construction site images, the model achieved a mean average precision at IoU 0.50 (mAP50) of 98.74% and a recall of 54.55%. While recall indicates room for improvement, the high precision highlights the practical value of synthetic data in enhancing model performance without requiring extensive field data collection. This research contributes a scalable approach to data generation in safety-critical and data-deficient environments, reducing dependence on direct data acquisition, while maintaining model efficacy. It provides a foundation for accelerating the deployment of AI technologies in high-risk industries by overcoming data bottlenecks and supporting real-world applications through practical synthetic augmentation.



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1. Introduction

Construction automation and robotics technologies play a crucial role in enhancing work efficiency and safety at construction sites. AI-based automation systems, including construction robots, drones, and automated inspection systems, require accurate object recognition, which relies on machine learning and deep learning models trained with large-scale, high-quality datasets [1]. In the construction industry, research is being conducted to manage sites using object recognition and model training with large datasets. Rho et al. [2] proposed a method that utilizes object recognition technology to automatically identify work targets and site conditions from worker-captured images, integrating this information with building information modeling (BIM) for construction process management. Jung et al. [3] developed a model that detects concrete cracks and extracts crack characteristics using a deep learning model trained on collected datasets. Considering the importance of data in the construction field, Park et al. [4] conducted a study on labeling objects in construction site images and classifying them by work type using a trained model.

Despite the increasing use of training data in research, constructing high-quality datasets for construction sites remains a significant challenge. Construction sites are complex environments with various structural elements, dynamic work conditions, and continuous changes. Large volumes of training data are required for accurate object recognition, especially when background information is diverse or objects have intricate shapes. In such cases, extensive labeled datasets are needed, making it difficult to effectively learn object features [5]. This issue is particularly evident in construction sites, where linear structures, such as scaffolding, are abundant, and background information is highly entangled. These characteristics make it difficult to produce reliable results using existing datasets. Furthermore, access to construction sites is often restricted, and there are safety risks during data collection. Construction sites are hazardous environments with heavy machinery and numerous workers, making it challenging to collect images from diverse angles and obtain a comprehensive dataset of various objects. Additionally, data collection incurs significant time and labor costs. The data shortage caused by these site-specific challenges is a major obstacle to the advancement of construction automation and robotics technologies. It directly affects model accuracy and limits real-world applicability.

To address these challenges, recent studies have aimed to improve object recognition performance by increasing the volume of training datasets through data augmentation and synthetic data generation. However, most existing augmentation methods are based on generic image processing techniques and fail to adequately reflect the unique characteristics and complexity of construction environments. For example, simple transformations, such as image rotation or color adjustment, have limitations in reproducing the realism of object placement and scene composition specific to construction sites. While 3D simulation-based synthesis methods are capable of generating photorealistic images, they require advanced technical expertise and substantial computational resources, and the modeling and rendering processes are complex and time consuming. GAN-based generation methods have the potential to produce novel visual content, but they offer limited controllability over the outputs, suffer from training instability, and face challenges in domain generalization, making them difficult to apply in real-world construction scenarios. In addition, the process of labeling large volumes of generated synthetic data is highly labor-intensive.

The most critical factor in generating synthetic data is ensuring that it closely resembles real-world environments. One of the most challenging aspects of data synthesis is adjusting object positions and sizes realistically. This issue can be addressed through augmentation and synthesis techniques. By utilizing prior knowledge of object types and placements, new data can be generated, while maintaining appropriate sizes and ground alignments. This allows for the addition or replacement of identical or different object types, ensuring realistic positioning. This approach enables efficient training dataset creation with minimal manual intervention. Additionally, it allows for the synthesis of various object types that blend naturally with their backgrounds, significantly expanding the range of object variations in the dataset. In addition, training datasets can be efficiently constructed without the need for complex labeling processes by leveraging information from existing open datasets. The ORES method proposed in this study offers several key advantages:

- Realistic data generation: objects and backgrounds are separated, and augmentation and synthesis are performed, while maintaining realistic object placement and size.
- Consideration of complex background elements: the augmentation process takes into account the presence of frequently occurring linear structures in construction environments, such as scaffolding.
- High applicability in construction automation: the generated dataset has strong potential for use in construction robots, automated inspection systems, and AI-based construction monitoring.

This study presents a cost-effective and high-efficiency data augmentation methodology applicable to construction robots and automation systems. By addressing the issue of insufficient training data, which significantly impacts AI system performance in construction sites, this method enables the creation of large-scale, realistic training datasets. This, in turn, helps overcome data shortages for construction robots and automated quality inspection systems. The contributions of this research include improving AI utilization in construction sites, establishing reliable datasets, and enhancing object recognition performance. Furthermore, beyond dataset construction for complex construction environments, this study is expected to expand the applicability of AI systems in similarly intricate backgrounds.

To achieve our objectives, the research process can be summarized as follows. In Section 2, the considerations and requirements for constructing a training dataset are first defined. The methods proposed in previous studies for constructing such datasets are analyzed and compared, and a design approach is presented. In Section 3, object range expansion synthesis (ORES) for the construction of a training dataset using synthetic data is introduced. The application of ORES is explained, targeting scaffolding (which accounts for a considerable proportion of falling accidents and can have a diverse background information) as a linear object. In Section 4, the performance of the proposed method is empirically validated.

2. Literature Review

Several key considerations must be taken into account when constructing a training dataset for accurate object recognition. First, ethical standards must be followed to ensure that data collection methods do not expose personal information. When collecting real-world data, personally identifiable information may be inadvertently included. If such data are collected, additional post-processing is required to remove or anonymize sensitive content, which results in increased labor and time. In this regard, synthetic data generation can offer a more efficient and privacy conscious alternative. Second, both the quantity and quality of training data are crucial. A large dataset with poor quality or a small dataset with high quality may both hinder the performance of object recognition models. Inadequate or imbalanced datasets can lead to performance degradation [6]. Therefore, the effectiveness of machine learning models largely depends on the sufficiency and quality of the data. A synthetic data generation approach that improves both the quantity and quality of training data, while minimizing time and labor costs is necessary. Third, accurate and consistent object labeling is essential for ensuring the reliability of the dataset. Object labeling is a critical step in preparing data for object recognition, yet it is often challenging to label newly collected data with high precision. Thus, it is important to develop high-efficiency synthetic data generation methods that maximize the use of existing open datasets with a small amount of labeled data. Finally, a model validation process using the constructed training dataset is required. This step should include quantitative performance metrics to verify whether the dataset contributes to improved object recognition accuracy.

The methods used to construct a training dataset can be distinguished based on the type of data used (Figure 1). The training dataset is divided into real and augmented datasets. Real datasets can be collected through visual devices (such as CCTV) or constructed via web crawling. However, collecting real data in construction sites can be challenging for the following reasons:

- **Environmental changes:** The environment at construction sites can vary considerably with the weather, season, and time of day, and these changes can compromise data consistency.

- Safety and accessibility: Owing to safety concerns, data collection may be restricted to certain areas or certain times. Additionally, many construction sites can be difficult to access, limiting the scope of data collection.
- Lack of diversity: Data collection can be limited to specific projects or sites that may struggle to achieve diversity. This can limit the generalizability of the model.
- Labeling challenges: Labeling data from construction sites can be a time-consuming process that requires expertise, making the accurate labeling of large volumes of data challenging.

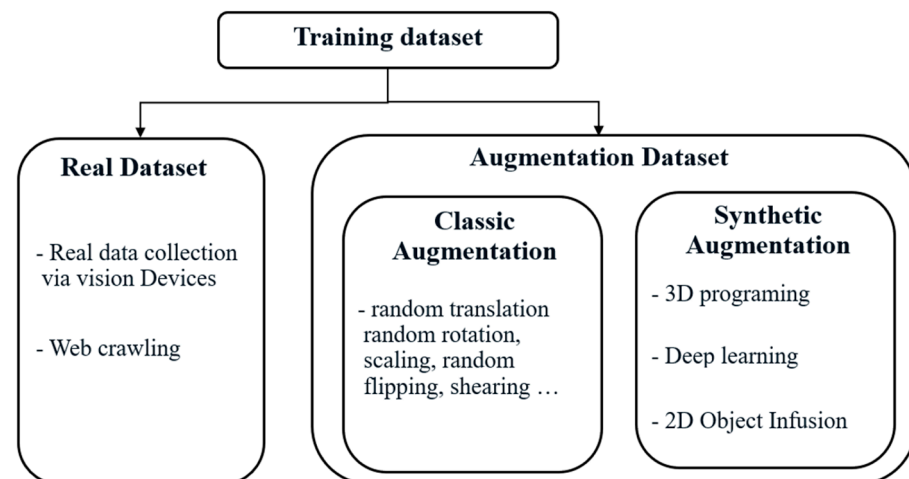


Figure 1. Training dataset building methods.

To overcome these limitations, the augmentation methods used for datasets can be divided into classic augmentation—which geometrically transforms existing training samples to increase the input diversity—and synthetic augmentation—which uses trained generative models on existing input training samples to create new, previously unseen samples [7]. Augmentation using synthetic data can be further categorized into methods that generate data through deep learning, methods that create synthetic data by modeling in virtual environments using 3D programming, and methods that insert objects into 2D images.

The requirements for generating synthetic data are summarized in Table 1. Utilizing large-scale synthetic data can help mitigate the degradation of data quality; therefore, it is essential to secure a sufficient volume of data through synthetic data generation methods [8]. In addition, the generated synthetic data must ensure diversity to produce realistic images [8]. When synthesizing various objects and backgrounds, environmental factors such as lighting, weather conditions, and perspectives must be taken into account, and a wide range of object types relevant to the construction domain should be included. Similarly, backgrounds should be composed of images that reflect diverse construction environments under different conditions. To ensure realism, the size and scale of objects in synthetic images must be appropriately adjusted to fit the surrounding environment, and the spatial relationship between objects and backgrounds should appear natural and realistic. Inserted objects should be resized according to the spatial context of the background and placed in a visually coherent manner in terms of position, depth, and perspective. If the size, scale, or positioning of objects in synthetic images does not align with the environmental context, discrepancies from real-world data may occur. For example, synthesized images showing scaffolding floating in the air or placed in construction sites where scaffolding is not needed can significantly differ from real data, ultimately degrading the performance of object recognition model training. Furthermore, lighting, shadows, and geometric consistency between the object and the background must be maintained to simulate the visual integration seen in real-world imagery.

Table 1. Requirements for generating synthetic data at construction sites.

Requirements		Detail
Quantity	- Data Volume	- A sufficient amount of training data must be available to ensure effective learning of the object recognition model. - The synthesis of diverse objects and backgrounds must consider changes in environmental conditions.
	- Object Variety	- Synthesized datasets must include a wide range of object types relevant to the construction domain.
Quality	- Background Variety	- Backgrounds should represent various construction environments, including different lighting, weather conditions, and perspectives.
	Diversity	- In the synthesized images, the size and scale of objects should be suitably adjusted to fit the environment.
	- Environmental Changes	- The synthesized object's relationship with the background should appear natural, and its placement should realistically align with the environment.
	- Scale Matching	- Inserted objects must be resized appropriately to match the spatial context of the background.
	Reality	- Object placement must appear natural in relation to the background in terms of position, depth, and perspective.
	- Scene Integration	- The lighting, shadows, and geometry between objects and backgrounds must be visually consistent to simulate real-world integration.

A comparison and analysis of the methods presented in 25 prior studies related to the construction of training datasets for construction sites, based on the previously defined requirements, is summarized in Table 2. Earlier studies that constructed training datasets using real data typically employed direct on-site data collection or web-crawling techniques. However, these methods face several limitations due to the temporary and dynamic nature of construction sites, making it both costly and labor-intensive to obtain high-quality training data. Furthermore, due to the continuous changes in construction projects, it is difficult to build generalized datasets from a limited number of sites. Data sharing is also hindered by issues related to privacy and confidentiality [9]. To overcome these limitations, research has shifted toward the use of augmented data. Augmentation methods are generally classified into classical and synthetic techniques. Classical augmentation focuses on modifying pixel values; although such techniques can be applied quickly, they provide limited diversity and fail to generate substantially new visual content. As a result, they are insufficient for capturing the complex visual characteristics of construction environments [10].

Table 2. Training dataset literature review by building method.

Method			Description Based on Requirements	Authors
Real Data	Actual field collection		Data collected from actual sites has high realism, but collecting a diverse set of data requires a lot of labor.	[9,11–13]
	Web crawling		The data collected in practice has high realism, but there are limitations in obtaining the desired variety of data.	[14]
Augmented Data	Classic augmentation	Simple enhancement through pixel value modification	It is possible to increase the amount of training data with less labor, but there are limitations in enhancing the diversity of backgrounds and objects.	[15–18]

Table 2. Cont.

Method		Description Based on Requirements	Authors
Augmented Data	3D rendering	Implementing the backgrounds and objects of construction sites in 3D requires a lot of labor and has lower realism compared to actual sites.	[19–21]
	GAN	GAN-generated images can be realistic, but building GAN models is labor-intensive and controlling the generated images is challenging.	[22,23]
	Generative AI	Pre-trained text to image models can efficiently generate photorealistic images from textual prompts. While less labor intensive the output quality and domain relevance rely on prompt design and precise control over image details remains limited.	[24–27]
	Object infusion	Enhances diversity by inserting objects and backgrounds, making it efficient for creating large datasets.	[28]

To address this, synthetic data generation approaches have been proposed to expand both the volume and diversity of training datasets. These can be broadly categorized into three types: methods based on 3D environments, those utilizing 2D environments, and hybrid approaches. While 3D modeling can simulate realistic spatial arrangements, it is generally limited to pre-modeled construction objects, and the creation of diverse and realistic scenes is time-consuming and resource-intensive. Moreover, adjusting parameters such as lighting, color, and material textures to reflect the variability of construction elements requires expert knowledge and adds further complexity [1]. In contrast, 2D synthetic approaches involve deep learning techniques—such as generative adversarial networks (GANs)—or object insertion strategies, where isolated construction objects are placed into real background images to create diverse training data. These methods are relatively more effective in achieving object-level diversity and visual realism with lower costs and simpler infrastructure. GANs are designed to learn the data distribution of real-world images in order to generate synthetic samples for realistic applications [29]. While GANs are effective at improving visual fidelity by minimizing unnatural artifacts, their application to construction site data synthesis remains limited. This is due to the inherent complexity, heterogeneity, and temporal variability of construction environments, which make it challenging for GANs to replicate the fine-grained spatial and contextual relationships required for domain-specific realism from the initial stage of generation. Recently, with advances in generative AI, new approaches such as text-to-image models and diffusion-based architecture have been explored for synthetic data creation. These methods offer the advantage of generating photorealistic images solely from textual prompts. However, their effectiveness is heavily dependent on prompt quality and design. Moreover, precise control over detailed visual elements remains difficult, which limits their applicability in structured and safety-critical domains, such as construction sites.

Table 3 presents a comparative analysis of synthetic data generation methods. Based on the comparison of synthetic data generation methods, object range expansion synthesis (ORES) demonstrates high practicality and resource efficiency. It can be implemented with minimal preprocessing using open datasets, supports batch processing, and operates without the need for high-end hardware. In contrast, methods such as 3D rendering, GANs, and generative AI (diffusion models, text-to-image) offer advantages in visual realism and expressiveness but require significant technical expertise, computational resources, and complex data preparation processes. Therefore, ORES is well-suited for large-scale training

dataset generation in the construction domain, providing a balanced solution in terms of accessibility, cost-effectiveness, and implementation efficiency.

Table 3. Comparative analysis of synthetic data generation methods.

Criteria	ORES (Object Range Expansion Synthesis)	3D Rendering	GANs	Stable Diffusion, Text-to-Image
Labor Requirements	Utilizes 2D datasets and basic image processing tools	Requires domain expertise in 3D modeling, simulation management	Involve deep learning model design, tuning, and implementation expertise	Requires prompt engineering and basic understanding of model behavior
Data Preparation Time	Open datasets can be used with minimal preprocessing	Preparation includes 3D asset creation, lighting setup	Require curation and annotation of training datasets	Preparation mainly involves designing and refining prompts; optional post-processing may be added
Data Synthesis Time	Supports fast batch processing; parallel generation achievable with low computing resources	Each scene must be rendered individually, which is computationally intensive	Generation time depends on training stability and convergence	Typically, within seconds to minutes
Cost Efficiency	Requires no specialized hardware or licensed software	Involves substantial investment in 3D modeling tools, rendering software, and computer infrastructure	Significant GPU resources and training time for model development	Can utilize publicly available pretrained models; large-scale deployment may still require GPU resources

In this study, we compared dataset expansion methods for object synthesis, as shown in Table 3. The object infusion using rendering method enables the rapid collection of objects from various angles within a 3D environment; however, when these objects are composited into a 2D background, aligning realistic angles between the background and objects proves challenging, resulting in low realism. In contrast, the object infusion using deep learning method can achieve high realism through deep learning processes, but it requires significant time and labor costs to implement the deep learning models. Meanwhile, the object range expansion synthesis (ORES) method leverages existing datasets to efficiently generate large-scale, diverse compositions, achieving a balance between low cost and time efficiency. This method also maintains adequate realism by synthesizing objects in appropriate spatial contexts. Therefore, the proposed object range expansion synthesis (ORES) approach provides high efficiency for large-scale dataset generation, while preserving a reasonable level of realism, making it cost- and time-effective for extensive data augmentation.

Table 4 presents a practical comparison between the proposed ORES method and existing synthetic data generation techniques, including 3D modeling, GANs, and generative AI. Leveraging open datasets, ORES enables rapid data preparation without requiring high-performance hardware and allows for scalable expansion through template-based insertion. Although it may lack algorithmic novelty, ORES demonstrates strong potential in terms of practicality, cost-effectiveness, and applicability in real-world construction environments.

Table 5 presents a literature review of object infusion methods used for data synthesis. It compares traditional rendering-based approaches, deep learning-based techniques, and the proposed Object Range Expansion Synthesis (ORES), highlighting their characteristics in terms of realism, efficiency, and resource requirements.

Table 4. Comparison between ORES and existing synthetic data generation methods.

Criteria	ORES (Object Range Expansion Synthesis)	Existing Methods (3D Modeling/GANs/Generative AI)
Data Preparation Time	Utilizes open datasets; minimal preprocessing enables rapid implementation	Requires time-consuming preparation (3D modeling, prompt engineering, data cleaning)
Hardware Requirements	Can run on standard PCs; supports parallel processing without high-end GPUs	Demands high-performance GPUs and large-scale computation resources
Design Complexity	Simplified design with fixed object position and scale	GANs suffer from instability; generative AI highly sensitive to prompt quality
Visual Coherence	Maintains object size, position, and background consistency for realistic images	3D rendering requires complex light/material tuning; GANs often lack background realism
Scalability	Easily extensible via template-based insertion across object and background types	Domain-specific retraining or model redesign required
Practical Usability	Applicable in short-term and small-to-medium-scale projects	Typically, requires large research teams and advanced infrastructure

Table 5. Literature review about object infusion.

Method	Description Based on Requirements	Features
Object infusion using rendering	3D-rendered objects can be inserted into 2D backgrounds to capture various angles, but realism in placement and angles is low	Allows for multiple angles but lacks realism, often resulting in unrealistic placements
Object infusion using deep learning	Generates realistic data using deep learning algorithms but requires significant time and effort to develop	Provides high realism but demands substantial resources and time
Object range expansion synthesis (ORES)	Matches various objects and backgrounds from an open dataset to synthesize new objects into existing images, creating diverse variations	Leverages open datasets to automatically generate diverse variations, ensuring high efficiency in data creation

Table 6 presents a comparison between the method described in this study and 15 other studies that proposed methods for generating synthetic data with respect to the construction of the training dataset. A comparison was made in terms of the quantity, quality, and efficiency of synthetic data creation, and a relative evaluation was conducted owing to the limitations of comparing absolute numbers. Neuhausen [20] generated data in a 3D environment to improve the recognition performance of workers in hazardous situations at construction sites using a cycle-rendering engine. Kim [10] proposed a virtual synthetic image creation method that overlaid worker objects created through 3D modeling onto real site images to address the lack of training data at construction sites. Zeng [17] augmented construction site images by modifying the image pixel intensity to address the effects of lighting changes at outdoor construction sites. Bang [30] used GAN-based image inpainting and the concepts of object class probability and relative size to remove construction objects from the site images obtained using unmanned aerial vehicles (UAVs) and reconstructed the removed areas with new objects. Consequently, efforts have been made to address the challenges associated with field data collection by synthesizing data to create training datasets.

Table 6. Synthetic data generation literature review.

Authors	Method	Quantity	Environment and Object
[1]	Framework for automatically generating a training dataset by creating images and labeling target objects using web crawling and virtual reality technologies.	Web crawling + synthetic image dataset and training dataset comprising 99,800 images in 42 min	Construction site and heavy equipment
[10]	Insertion of workers in various poses modeled in 3D into 2D construction site backgrounds, resulting in a 50% reduction in the actual data needed.	Hybrid dataset 10,000 (worker) (Real5000 + Syn5000)	Construction site and worker
[14]	Automatic collection of desired images from the Internet, reflecting various visual characteristics of objects (same object, different manufacturers, etc.) for training images. Automatic image labeling using an image segmentation model. Completely random cross-over sampling of the foreground and background.	Automatically creates a training dataset comprising 5864 images in 53.5 min	Construction site
[19]	Generating 2D synthetic images of building elements (e.g., columns, walls, and slabs) using building information modeling (BIM). Synthetic datasets are used to train dense neural network (DNN) architectures. The developed model effectively and accurately positions building elements in real construction images.	-	Building
[20]	Data generation in a 3D environment to improve the recognition performance of workers in hazardous situations on construction sites. Rendering using the cycles rendering engine, a mini batch size of 64 based on the Darknet 53 model (pretrained on ImageNet).	8 different scenes Total 3835 frames 32 tracked subjects	Construction site and worker
[21]	Generating a large-scale labeled dataset for excavator pose estimation using domain randomization with a gaming engine.	12,000 synthetic images (training dataset) 3000 real images (validation dataset)	Construction site and excavator
[23]	Creation of synthetic images through the removal of workers from construction site images and inpainting using a U-Net model optimized using Adam. Applying context encoders to remove duplicated objects in images and inpainted background context, with areas requiring context inpainting being predefined in size and location within the image. Application of the U-Net deep learning architecture for direct image-to-image conversion to relax the fixed size and location constraints of context inpainting.	5846 construction images	Panoramic view of a construction site where objects have been removed
[25]	Synthetic images were generated using stable diffusion with construction-related prompts. Context-based labeling was applied to enhance dataset quality.	150,000	Construction task, CNN trained with context-based labeling
[26]	A text-to-image model was used to generate images of 27 construction hazard scenarios based on structured prompts.	3585 images across 27 scenarios	Construction accidents, object relations captured

Table 6. Cont.

Authors	Method	Quantity	Environment and Object
[27]	Pretrained generative AI was used to create scaffolding images. The model learned distinctive features, though image diversity was limited.	-	Scaffolding, segmentation performance evaluated
[30]	Generation by combining 3D models of construction equipment and various background images taken at construction sites. Training of a machine learning-based excavator detection model using only synthetic datasets. Improving detection accuracy while reducing the time required for image annotation.	3D model with 16 backgrounds 765 (no. of positive images)	Construction site and excavator (modeling)
[31]	Development of a framework for object recognition during curtain wall installation work from tower cranes. Insertion of labeled images of curtain wall panels and crane hooks created with the Unity engine. Creation of training data and understanding crane movements using the intersection over union (IoU) tracker.	Generate 300,000 training images per hour	Curtain wall work site via crane and crane hook and curtain wall panel
[32]	Building an excavator database using 3D modeling tools and game engines. Building a synthetic image database including the location and pose of excavators.	-	-
[33]	Building a training dataset by generating dust in a 3D environment and inserting data created in a construction background with inserted dirt.	3860 synthetic dust images (training dataset) 1015 real images (test dataset)	Construction site with dust inserted
[34]	Building virtual data from real building images through BIM and augmented data using a GAN.		Inside the building
[35]	Data collection using UAVs. Use of GAN-based image inpainting and concepts of object class probability and relative size. Method of removing objects (e.g., excavators) from site images and reconstructing new objects (e.g., mobile cranes) in areas where objects have been removed.	544 (training dataset) 112 (test dataset)	Construction site and heavy equipment (taken by UAV)
[36]	Photorealistic images of construction accidents were synthesized using stable diffusion. Prompts were crafted to reflect real-world safety incidents.	2324 fall accident cases analyzed using GPT-4, 300 synthetic images generated	Safety incidents, object detection, and action recognition
Our re-search	Efficiently generates high-quality synthetic data by separating backgrounds and objects and inpainting. Allows realistic synthesis of various object forms in original locations, expanding object range and diversity with minimal real data and without extra labeling or 3D modeling.	Generate 30,000 synthetic images	Scaffolding Worker ladder

To better situate our contribution in relation to existing methods, we provide an analytical synthesis of the techniques summarized in Table 6. Prior studies have explored various synthetic data generation strategies using 3D rendering, domain randomization, GANs, and generative AI techniques. Many approaches focused on specific objects, such as excavators [21,30], workers [10,20], or safety incidents [25–27,36], often relying on resource-intensive processes, such as 3D modeling [19,24], unity-based simulations [23], or text-to-

image diffusion models [36]. While these methods demonstrated strong visual realism or automation potential, they typically require significant time, expertise, or hardware resources, limiting their short-term field applicability. In contrast, the proposed ORES method prioritizes efficiency and real-world usability, offering a lightweight, easily scalable alternative without the need for high-performance infrastructure. Unlike most prior work, our study specifically targets linear and safety-critical objects (scaffolding, ladders), which are underrepresented in existing literature despite their high relevance to fall-related construction accidents. By minimizing preprocessing and eliminating the need for 3D modeling or manual labeling, ORES enables the rapid construction of realistic, diverse datasets supporting practical deployment in safety-focused AI systems with lower operational burden.

Considering their diversity of backgrounds and objects, construction sites require various arrays of objects and backgrounds in their training data. Additionally, owing to the frequent changes in object types and visual characteristics at sites depending on the construction work, there is a need for the ongoing development of the training dataset [1]. However, previous studies that generated training datasets using synthetic data primarily focused on relatively easy-to-synthesize objects, such as heavy machinery and workers, which limits their applicability when the types of objects vary. Additionally, creating synthetic data that considers realistic positions and scales is challenging, and the labor required to achieve realistic placement is substantial. Consequently, a method is required that can secure training data for a range of objects and efficiently generate substantial volumes of them.

3. Methods: Object Range Expansion Synthesis (ORES)

3.1. Distinction of the Proposed Method

When generating synthetic data using 3D-modeled objects, it is often necessary to remodel the object or reconstruct the 3D environment if the object type changes. This study proposes a method to increase the volume of data by leveraging existing open datasets and a small amount of training data, enhancing the diversity of synthetic data through the separation and synthesis of objects and backgrounds. The proposed method is distinct in that it allows for the construction of training datasets using a small dataset without additional labeling, even when the object-type changes. Additionally, it is possible to synthesize objects of the same type or different types by aligning them to realistic positions and scales using existing information about object locations and types. Table 6 presents an analysis comparing previous studies with the current research. Through the ORES method proposed in this study, it is possible to synthesize both planar objects, such as workers with a solid background, and linear objects, like frame scaffolds, which have open areas in the background. Additionally, using object and background information from existing open datasets, object synthesis can be performed in a way that aligns with the background context. For example, if background data from masonry and plastering work is available, objects like ladders or workers, which are likely to be found in this construction type, can be synthesized instead of frame scaffolds, thereby generating a more diverse dataset. The proposed object range expansion synthesis (ORES) method distinguishes itself by enabling the expansion of object ranges to create diverse datasets using only a small amount of field data and leveraging background information. Moreover, since the open datasets used already contain labeled information, there is no need for additional labeling, thereby minimizing labor and time costs.

Table 7 presents analysis of previous studies and distinctiveness of this research. Each method is evaluated in terms of its limitations, object types, ability to specify realistic compositing locations, and automation of annotation. While traditional approaches often suffer from unrealistic placement and limited background context integration, the proposed

Object Range Expansion Synthesis method demonstrates improved accuracy on object placement by leveraging existing location information, and supports automated annotation, ensuring synthetic dataset generation for construction environments.

Table 7. Analysis of previous studies and distinctiveness of this research.

Method	Ref. No.	Limitations	Synthetic Data Generation				
			Object		Specify Realistic Compositing Location	Annotation Automation	Object Range Expansion Synthesis
			Paner	Linear			
3D Modeling + Object Infusion	[1]	Unnaturalness from web-crawled objects Including background	Heavy equipment	✗	△ Random located except upper part	✗	✗
	[10]	Modeled worker differs from real images.	Worker	✗	✗	○	✗ New 3D model creation required
	[30]	Synthetic images created with random backgrounds differ from real images.	Excavator	✗	✗ Located without considering background	○	✗
Object Infusion	[14]	Synthetic images do not account for ground conditions and environments and construction sites.	Excavators	✗	✗ Random located	✗	✗
	[24]	Image quality is highly dependent on prompt design, making detailed control difficult.	Small construction tools (bucket, cord reel, hammer, tacker)	✗	△	✗	✗
	[35]	Discrepancy between the remaining original background and the composite background of the separated object	Heavy equipment	✗	△ Approximate positioning of the composite		✗
Object Range Expansion Synthesis (Our research)		Limited in handling dynamic environmental factors such as lighting changes, occlusion	Worker	Frame scaffolding, ladder, mobile scaffolding	○ Accurate placement based on existing object location information	○	○

○: Applicable, ✗: Not Applicable, △: Partially Applicable.

Training datasets collected from actual field data are limited in quantity and diversity due to restricted objects and environments within the same space. Collecting and labeling large, diverse datasets manually is an extremely time-consuming task, which can delay the deployment of computer vision technology in applications such as construction equipment recognition [30]. Furthermore, the process of creating synthetic data also involves labor-intensive data labeling. Therefore, this study aims to address these issues by proposing a method that reduces labeling labor, while improving object recognition rates using a small amount of data.

Figure 2 illustrates the number of possible cases for creating synthetic data by separating objects from the original image and performing inpainting. In the proposed ORES method of this study, object replacement involves generating data by replacing the object in its original position with either a different instance of the same object type or an entirely different object. In Figure 2 below, objects were first separated from the original image containing a scaffold and a worker, followed by inpainting. The first synthetic image shows the result of combining a different object of the same type. The second synthetic result

illustrates the combination of a different object type, where a ladder was placed in the position of the scaffold, and a ladder was combined with a worker. Another example shows a replacement of the scaffold with a trestle scaffold. As such, this study demonstrates that the required objects for training datasets can be generated in large quantities using various backgrounds and objects from open datasets. Furthermore, depending on the variety of objects available, it is possible to significantly expand the scope of objects and construct richer datasets.

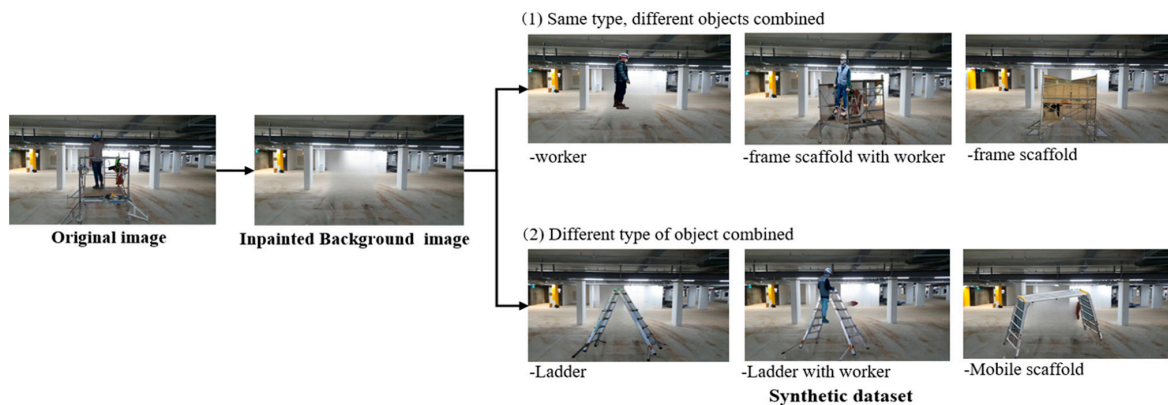


Figure 2. Synthetic dataset generation via ORES.

3.1.1. In Terms of Quantity and Variety

Combining classic and synthetic augmentation can enhance the variety of training data, positively impacting model performance. As shown in Figure 3, a limited number of images can be divided into backgrounds and objects, each of which can be separately augmented to increase data quantity. Without augmentation, the single image remains unchanged. However, by independently augmenting the backgrounds and objects, their volume can be effectively doubled. Subsequently, combining the augmented objects and backgrounds results in a notable increase in the quantity and diversity of the training data. However, repeated training on the same data runs the risk of overfitting, potentially reducing the accuracy and generalizability of the model. To mitigate the risk of overfitting and ensure diversity, different types of objects and their augmented datasets can be randomly mixed during training. By using labeled 2D objects from one image, we can place these objects on various background images. This method of data synthesis introduces domain randomization, which enhances both the data diversity and generalization capabilities, as different objects are synthesized into new images. Although generating images that perfectly resemble real images can be challenging, expanding the dataset with slightly lower-quality synthetic data can help reduce overfitting and improve both the accuracy and generalization abilities.

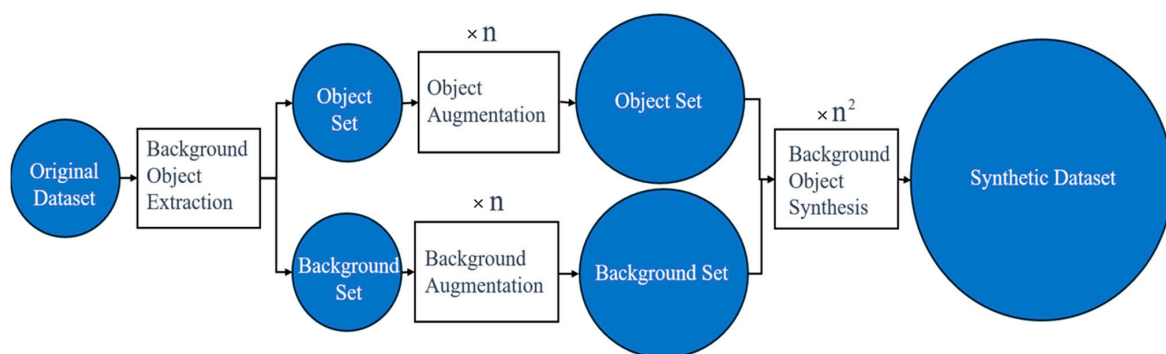


Figure 3. Concept of introducing diversity in training data.

Consequently, the data synthesis method proposed in this study efficiently expands a small number of images into a larger dataset, ensuring diversity, preventing overfitting, and facilitating the construction of an optimal training dataset.

3.1.2. In Terms of Quality

Using object insertion as a synthesis method is more labor efficient than modeling in a 3D environment. It also provides faster synthesis and enables the insertion of various object types within the same timeframe. Traditional object insertion synthesis often involves the random or manual placement of objects onto the background. However, random placement can result in synthetic data with unrealistic object positions and scenarios, and random or inappropriate augmentation can potentially diminish model performance [7]. Bang et al. [35] generated synthetic data with low realism by placing construction objects in image gaps, without considering the actual terrain or construction site conditions. This method has limitations, as it introduces object placements that do not exist in real construction sites, which can degrade the recognition performance.

The proposed method separates objects from the background and retains information on the types and positions of the separated objects. During the augmentation of different objects, contextual appropriateness is considered during the synthesis process. Additionally, the size of the original separated objects aids their placement in the background at an appropriate scale. This approach facilitates the construction of a synthesis dataset in which object sizes and types are contextually appropriate, in contrast to previous studies. Manual synthesis allows for human inspection and object placement; although, manually creating of a large volume of training data is time intensive.

Consequently, in this study, the synthesis process was automated, enabling users to just select the object types and their placements. This automation facilitates the efficient generation of synthesized images in the desired quantities, ensuring the efficient production of synthetic data that resemble real-world data.

3.2. Process of Building Synthetic Dataset

The synthetic dataset proposed in this study was constructed as follows: (1) image labeling and object separation from the background; (2) augmentation and synthesis of the background and object; and (3) object recognition model evaluation (Figure 4).

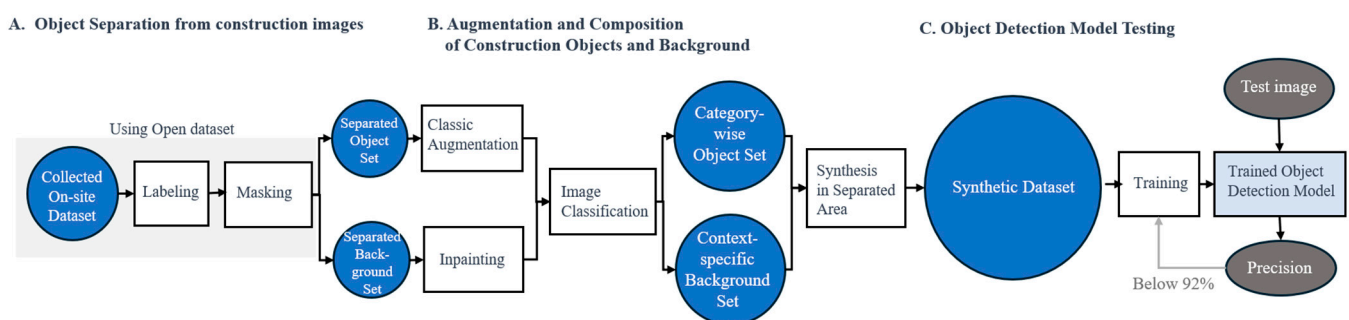


Figure 4. Object range expansion synthesis (ORES) process.

The process of applying the synthetic dataset construction, as defined in Figure 4, to create synthetic data for scaffolding is shown in Figure 5.

3.2.1. Image Labeling and Object Separation from Background (Figure 5A)

The code for generating the synthetic images was written in Python 3.9.19 using Visual Studio (<https://visualstudio.microsoft.com/>, accessed on 21 April 2025), and labeling was performed using LabelMe (<https://labelme.io/>, accessed on 21 April 2025). Additionally,

the data used to train the initial object recognition model were directly collected from the field, and the construction site image data were collected using AI-Hub (<https://www.aihub.or.kr>, accessed on 21 April 2025), a publicly available dataset platform for AI research In Korea.

Labeling was performed on the original scaffolding image (Figure 6) using segmentation. This process includes the separation of the background and object by applying the “you only look at coefficients” (YOLOACT) model to the labeled scaffolding field data. In this study, time was saved by using a prelabeled OpenAI annotation dataset. Labeling training data typically requires considerable time and effort owing to the manual labeling of large datasets. However, the time and labor costs can be minimized by initially labeling the objects intended for synthesis and subsequently synthesizing the labeled object data onto the background data. This method enables the quick and efficient creation of a training dataset, surpassing the traditional method of labeling all augmented and expanded datasets in terms of speed and efficiency.

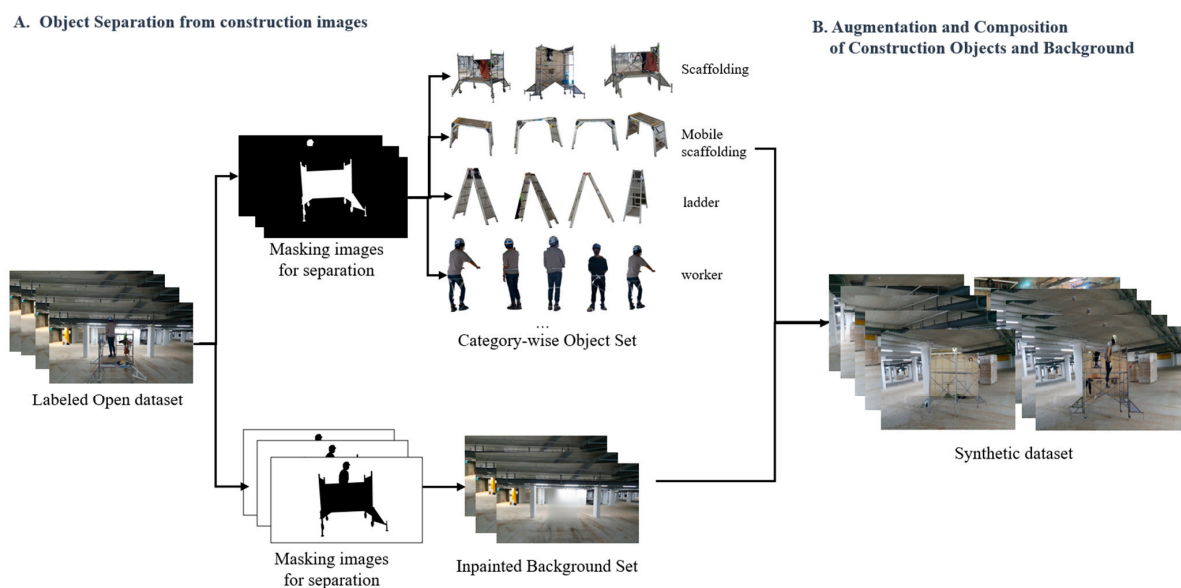


Figure 5. Synthetic data generation process of scaffolding based on the proposed method.



Figure 6. Original scaffolding images.

3.2.2. Augmentation and Synthesis of Background and Object (Figure 5B)

The separated object undergoes an augmentation process, and the background from which it is separated is painted. Objects extracted from construction site images can then be enhanced through classical augmentation (such as color, saturation, and brightness adjustments). In object augmentation, the input values can be modified between -255 and 255 using an algorithm. Concurrently, the backgrounds from which the objects are removed can be naturally processed through inpainting. Figure 7 shows the background image after the separation and inpainting of the scaffolding, and it illustrates the natural AI-based processing of the original scaffolding. The backgrounds of the separated objects can be categorized based on the field image context. The background dataset comprises both inpainted backgrounds from which objects are removed and backgrounds with no object separation.



Figure 7. Inpainted background image.

The subsequent step entails synthesizing the object dataset, sorted by type, with the background dataset categorized based on construction site conditions. This synthetic data creation technique involves isolating the existing objects and then synthesizing various forms of the same object type in their original locations. The type and size of the synthesized object can then be determined based on those of the original object. As the object's placement aligns with the background context, and its size is calibrated based on the original object's size, this method enables the generation of highly realistic synthetic data, mirroring the actual site conditions.

Figure 8 illustrates the synthetic scaffolding data generated using this method, showing multiple variations of the same object type synthesized into a single background. Figure 9 presents the results of same-type object synthesis and cross-type object synthesis, where objects are replaced with either the same type or a different type. This method addresses data diversity problems and expands the dataset by integrating various object forms into a single background. Moreover, a modest volume of initial training data can be transformed into an extensive dataset through object separation, augmentation, and background synthesis.

In this study, 4000 real data images were used to generate 30,000 synthetic images, and the creation of the 30,000 training data images was achieved in 30 min. This demonstrates

that a substantial volume of training data can be rapidly produced with a minimal set of real data using synthetic data techniques.



Figure 8. Scaffolding synthetic data.

Synthetic images



Figure 9. ORES-based synthetic data: examples of same-type and cross-type object replacement.

3.2.3. Test of Object Recognition Model

The final stage involved assessing the accuracy of the object recognition model using the synthesized images. If the accuracy fell below 92% after training using the synthetic data and testing with real data, the process reverted to the object recognition model training phase. Continuous training enhances the model's accuracy, leading to the development of an object recognition model with a high recognition rate.

4. Validation

4.1. Validation Scenario

Validation tests were conducted to evaluate the performance of the object recognition model trained using synthetic data. The dataset for assessing the object recognition accuracy comprised 30,000 synthetic images for training and 1500 real images for testing. Model training involved dividing the synthetic data into a training dataset of 21,000 images and validation dataset of 9000 images (in a 7:3 ratio). Owing to the large time and labor costs involved in obtaining an equivalent volume of real data as synthetic data in the field, tests were conducted using 1500 real data images. The training and development environment comprised the RTX4080 (Nvidia Corporation, Santa Clara, CA, USA), RTX3080 (Nvidia Corporation), Ubuntu 20.04 LTS, Pytorch 2.0.1, and ResNet50 as the backbone.

4.2. Results of Validation

The core contribution of this study lies in the application of a practical, domain-specific synthetic data strategy object range expansion synthesis (ORES) tailored for construction safety contexts. In particular, the study focuses on generating synthetic datasets for linear structures, such as frame scaffolds and ladders, which are frequently associated with fall-related accidents and represent a significant safety concern on construction sites. These objects are structurally sensitive to visual inconsistencies due to their transparency and the need for background continuity, making it difficult to ensure visual realism using conventional data augmentation or synthesis techniques. This challenge is further amplified in construction environments, where lighting, background, and spatial conditions are constantly changing.

Therefore, this study prioritizes practical applicability and domain relevance over purely technical novelty, selecting safety critical and contextually representative objects to validate the effectiveness of the proposed approach. Although the experimental scope was limited to a small number of object categories, the methodology demonstrates substantial academic value as a data-driven solution to real-world safety issues. We plan to expand the scope of this research by including additional objects and high-risk scenarios, such as rooftop work and aerial operations to further enhance its generalizability and field applicability. By focusing on difficult-to-represent safety-critical structures, this study demonstrates that domain-aware data synthesis can be both feasible and impactful for training AI models in complex real-world environments

Figure 10 shows the results of the object recognition model trained with 30,000 synthetic data images and tested on real data. The segments marked in red indicate the recognition of scaffolding, whereas those marked in blue indicate the recognition of workers. Training with synthetic data yields high recognition rates, even at long distances. Although precisely identifying the exact boundaries of the scaffolding proved to be challenging, recognizing the general shape of the scaffolding and workers using bounding boxes was feasible. When evaluating the performance of the recognition model, an improvement in accuracy with an increased number of epochs was evident when trained using 30,000 synthetic data images (Table 8). The intersection over union (IoU) threshold—which assesses the overlap between the predicted and actual object locations—was maintained at 0.5, following the recommendation of most prior studies. The recognition model's accuracy at this threshold is summarized in Table 9. By comparing the accuracy results with those of previous studies that used synthetic data for training, an enhancement in the performance in all aspects is evident (Table 10).



Figure 10. Scaffolding recognition results.

Table 8. Test results (mAP).

Epoch	Average	mAP50	mAP55	mAP60	mAP65	mAP70
10	35.55	94.68	92.39	85.34	70.18	32.47
50	43.84	98.62	97.60	97.54	92.04	68.50
100	44.9	98.63	98.63	97.33	94.46	72.51
150	44.29	98.74	98.74	98.36	92.93	71.05
200	44.61	98.70	98.70	98.35	93.22	71.90
250	44.41	98.74	98.63	98.29	92.92	71.28

Table 9. Accuracy of scaffolding recognition model trained using synthetic data.

Epoch	mAP50	F1score	Precision	Recall
250	98.74	64.59	99.24	54.55

However, despite the high precision observed, the recall remained relatively low (54.55%), indicating that some objects, particularly scaffolding structures, were not detected during testing. To further investigate and improve this limitation, we propose the following analyses and enhancement strategies.

- **Object-wise precision and recall analysis:** Additional object categories will be introduced to compare precision and recall across multiple object types. This will help determine whether the low recall is a consistent issue across all categories or primarily concentrated in specific object types.
- **Augmentation of complex visual conditions:** In the current approach, objects and backgrounds are separated and recombined to generate synthetic data. However, the training data lacks scenarios involving occlusion, partial visibility, and lighting variation. To address this, future work will include additional augmentation techniques that simulate changes in viewpoint, illumination, and background context. These enhancements are expected to improve recognition performance in challenging visual environments.
- **Integration of real and synthetic data:** To reduce the domain gap between synthetic and real-world data, a hybrid training strategy will be explored. By combining a proportion of real images with synthetic ones, it is expected that the model can benefit from the realism of actual data, while leveraging the scalability of synthetic data, thereby enhancing recall and overall robustness.

Table 10. Accuracy comparison with existing research using synthetic data.

Method	Dataset	Object	Algorithm	Precision	Recall	F1 Score
3D rendering [20]	8 different scenes Total 3835 frames 32 tracked subjects	Construction site and worker	CNN YOLOv3	94%	-	-
3D rendering + object infusion [1]	Web crawled + synthetic image database. Training database comprising 99,800 images in 42 min	Construction site and heavy equipment	YOLOv5 Batch size 16, Faster R-CNN	-	-	63.11%
3D rendering + object infusion [10]	Hybrid dataset 10,000 (worker) (real5000 + syn5000)	Construction site and worker	DNN YOLOv3	67.8%	-	-
3D rendering + object infusion [30]	3D model with 16 backgrounds 765 (no. of positive images)	Construction site and excavator (modeling)	-	75%	98%	-
Web crawling + object detection [14]	Automatically creates 5864 data (179 s)	Construction site and heavy equipment	Faster R-CNN	92.71%	88.14%	-
Deep learning + object infusion [35]	76,320 synthetic data (6 different construction sites)	Construction site and heavy equipment (taken by UAV)	Faster R-CNN	57.13%	60.22%,	-
Object range expansion synthesis (ORES)	30,000 synthetic data	Scaffolding	YOLACT	99.24%	54.55%	64.59%

5. Conclusions

This study proposed object range expansion synthesis (ORES) as a practical and efficient method for generating large-scale training datasets from limited real-world data. By inserting 2D objects into spatially annotated backgrounds, ORES enables rapid dataset generation without requiring 3D modeling or programming. The experimental results on scaffolding confirmed that the proposed method achieved an object recognition performance comparable to that of existing approaches. The proposed approach has strong potential for real-world applications in AI based construction automation systems, including object detection, quality inspection, autonomous equipment, and drone-based monitoring. Especially in environments where real data collection is difficult or hazardous, ORES offers a cost-effective and scalable solution for generating diverse and realistic training datasets.

To demonstrate the practical applicability of ORES, we present a deployment roadmap in Figure 11, outlining the complete pipeline from real-world data collection to on-site application. The process includes synthetic dataset generation via ORES, training of object detection models, and actual deployment in tasks such as on-site risk alerting, quality control monitoring, and drone-based safety surveillance. Notably, the workflow incorporates a feedback mechanism for continuous model enhancement based on missed detections or false positives. This practical roadmap bridges the gap between synthetic data generation and operational implementation, highlighting the field readiness of ORES in dynamic construction environments.

However, this study has several limitations. First, while the model achieved a high mAP50 of 98.74, the recall was relatively low at 54.55, indicating that some actual objects were not detected. This is likely due to the visual similarity between scaffolding and background elements, which hinders effective feature extraction. Second, the current im-

plementation lacks the precise consideration of viewpoint alignment between objects and backgrounds. Object and background angles were simplified into three categories—frontal, side, and oblique—but this classification is insufficient for accurate visual alignment. As a result, the synthesized images, particularly for linear objects (frame scaffold, ladder), often appear unnatural when the background behind the object differs visually from the insertion background. To address this, future work should consider a background-matching mechanism that selectively pairs backgrounds with similar color and environmental features, especially in cases where the background is partially visible through the object structure. Moreover, the current study focuses solely on scaffolding, limiting the generalizability of the results across broader construction scenarios.

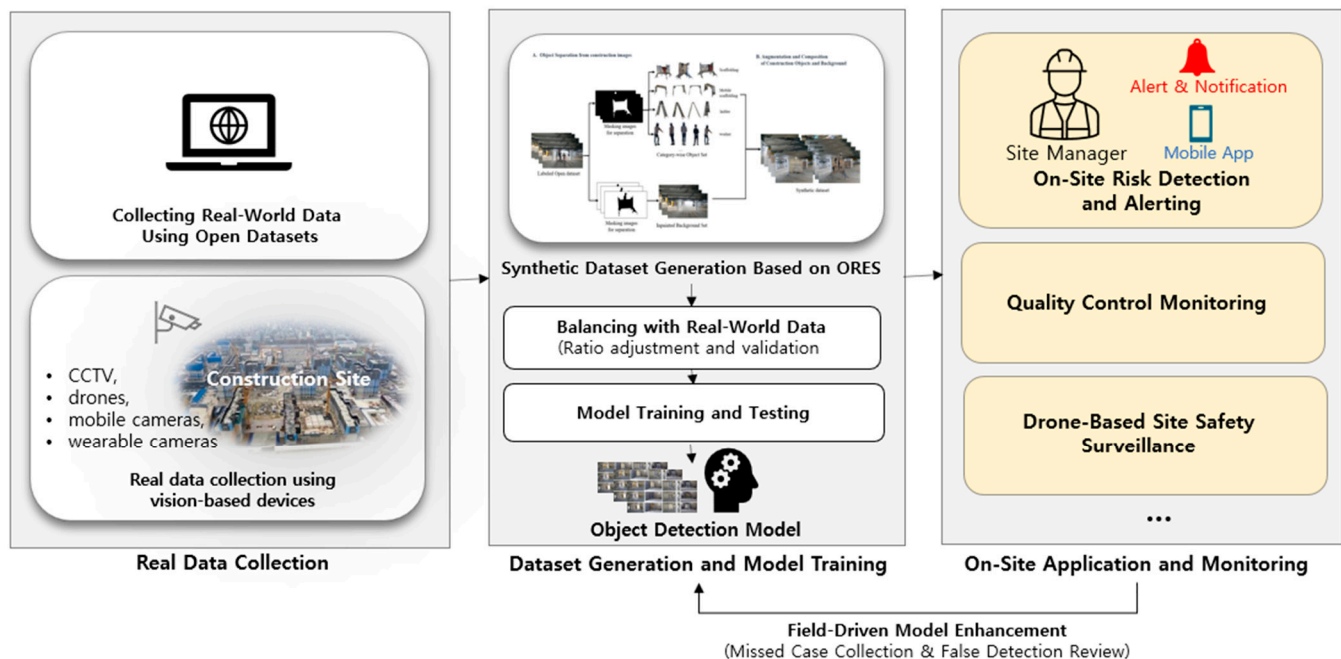


Figure 11. Workflow for real-world implementation of ORES in construction site applications.

To enhance the applicability and robustness of the proposed model, future research should pursue several key directions. First, object-wise precision and recall analysis should be conducted by expanding the variety of object types used in training and evaluation to determine whether the observed low recall is a general issue or limited to specific categories. Incorporating advanced augmentation techniques that simulate complex visual conditions, such as occlusion, partial visibility, lighting variation, and viewpoint diversity, will help improve recognition performance under real-world scenarios. It is also essential to expand the diversity of construction objects and environmental contexts included in training to ensure better model generalization. In addition, applying hybrid training strategies that combine synthetic and real-world data in balanced proportions is expected to reduce domain gaps and improve recall performance. Lastly, field-based experiments, such as crane automation and drone-based inspections, should be conducted to validate practical feasibility and the real-world performance of the proposed approach.

In conclusion, ORES presents a practical, field-oriented approach to addressing the issue of data scarcity in the construction domain. Its key strength lies in enabling the efficient and scalable generation of training datasets without the need for advanced technical infrastructure. The method is particularly suitable for safety-critical scenarios involving high-risk objects, such as scaffolding and ladders, and it can be effectively applied even in resource-constrained environments. By balancing realism, usability, and adaptability, ORES not only enhances object recognition accuracy but also offers a meaningful contribu-

tion toward the broader adoption of AI-based safety management systems in real-world construction settings. This balance positions ORES as a domain-aware, resource-efficient solution to advanced AI safety applications in construction.

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